Bahram Behzadian

Reinforcement Learning · Post-Training · Model-Based RL

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Research Statement

I work on reinforcement learning and sequential decision-making. The through-line across my research is policy improvement when the model of the world or the evaluator is imperfect: robust MDPs when the dynamics are uncertain, and truncated or approximate evaluators when the reward signal is unreliable. My recent work studies policy improvement under verifier-style rewards, connecting to post-training and RL for reasoning models. Earlier work spans robust MDPs, learning under uncertainty, and partial observability. I am increasingly focused on model-based RL and reasoning systems that plan under imperfect models.

Publications

UPI-TRM: Policy Improvement under Truncated Evaluators with Verifier-Style Rewards.
ICLR 2026 RSI Workshop; NeurIPS 2026 (under review).
Decomposes finite-depth evaluator error into a Bellman residual and a truncation bias, propagated through a conservative policy improvement bound. Connects to RLVR and verifier design.

Fast Algorithms for L_∞-Constrained S-Rectangular Robust MDPs.
Neural Information Processing Systems (NeurIPS), 2021.
Scalable algorithms for robust policy computation under structured uncertainty.

Optimizing Percentile Criterion using Robust MDPs.
International Conference on Artificial Intelligence and Statistics (AISTATS), 2021.
Robust decision-making under risk-sensitive and quantile-based objectives.

Fast Feature Selection for Linear Value Function Approximation.
International Conference on Automated Planning and Scheduling (ICAPS), 2019.
Representation and feature selection for value-based reinforcement learning.

Monte Carlo Localization in Hand-Drawn Maps.
IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2015.
State estimation under partial observability with noisy map representations.

Workshops and Other Venues

Optimizing Norm-Bounded Weighted Ambiguity Sets for Robust MDPs. NeurIPS Workshop on Safety and Robustness in Decision-Making, 2019.

Feature Selection by Singular Value Decomposition for Reinforcement Learning. ICML Workshop on Prediction and Generative Modeling, 2018.

Low-Rank Feature Selection for Reinforcement Learning. ISAIM, 2018.

Robot Navigation in Hand-Drawn Sketched Maps. European Conference on Mobile Robotics (ECMR), 2015.

Experience

2022–Present **Research Scientist**, *Meta*, Palo Alto, CA
- Build large-scale experimentation and evaluation infrastructure: metric correctness pipelines, anomaly and data-quality monitoring, and ML auditing for production systems.
- Design statistically grounded evaluation frameworks that surface when large-scale learning systems are unreliable, biased, or being gamed.

2018 **AI Engineer Intern**, *Envio, Inc.*, Dover, NH
- Learning-based methods for vehicle routing and scheduling as constrained sequential decision-making problems.
- Combinatorial planning and long-horizon trade-offs under execution uncertainty.

Education

2015–2022 **Ph.D. in Computer Science**, *University of New Hampshire*
Dissertation: *Efficient Data-Driven Robust Policies for Reinforcement Learning* (advisor: Marek Petrik). M.S. in Computer Science awarded en route (2018). Thesis: *Representation Learning for Reinforcement Learning via Singular Value Decomposition*.

2010–2013 **M.S. in Machine Automation**, *Tampere University*
Thesis: *Robot Localization under Partial Observability with Weak Maps*

Honors and Fellowships

CEPS Graduate Fellowship, University of New Hampshire

Thesis and Dissertation Fellowship, Tampere University of Technology

Skills

Reinforcement Learning: robust MDPs, learning under uncertainty, partial observability, model-based RL, planning, policy improvement under imperfect evaluators, verifier-style rewards and RLVR.

Post-training: DPO, PPO, GRPO, reward modeling, evaluation methodology.

Engineering: Python, PyTorch, large-scale experimentation and evaluation infrastructure.